Sound speed and resonance

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel time and explain when resonance occurs to interpret sound.

SUCCESS CRITERIA - I can...

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At fixed wave speed, frequency becomes **3** times as large. **Compare wavelength** and typical diffraction through the same opening.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $d_{echo} = v\,\Delta t/\mathbf{2}$; v = **340 m/s**.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

Worked reasoning: In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=**0.5 m** and sound speed **340 m/s**, f=**340/1**=**340 Hz**.

Your turn: Sketch the fundamental with displacement antinodes at both ends.

open-open tube

_____

_____

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An echo returns from a wall.
1. **Draw the outgoing** and returning path.
2. Why does sound's total travel distance exceed wall range?



Marzano check - circle: ① ② ③

04 Apply the physics

In air with sound speed **340 m/s**, an echo returns after **0.30 s**. **Find the wall distance**. In a separate practice example, an open-open pipe is **0.50 m** long. Use λ = **2L** and f = v/λ to **find its fundamental wavelength** and frequency.

Measurement	Path / wavelength	Result
Echo: 0.30 s	Out and back	____ m
Open pipe: 0.50 m	λ = 2L	____ Hz

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A heat engine absorbs Q_{hot}=**3500 J** and rejects Q_{cold}=**2450 J** each cycle. **Find work** and efficiency. **Label energy entering**, leaving and stored.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

A closed-open pipe fundamental has L=**0.25 m**. Use λ=**4 L** and v=**340** to **find frequency**.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.85 m** tube is closed at one end and open at the other.
1. Use v=**340 m/s** and λ=**4 L** for its fundamental.
2. **Find the first three allowed** ideal resonance frequencies.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3